

WAVES: Weighted Average of Signal Subspaces for Robust Wideband Direction Finding

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Abstract—Existing algorithms for wideband direction finding are mainly based on local approximations of the Gaussian log-likelihood around the true directions of arrival (DOAs), assuming negligible array calibration errors. Suboptimal and costly algorithms, such as classical or sequential beamforming, are required to initialize a local search that eventually furnishes DOA estimates. This multistage process may be nonrobust in the presence of even small errors in prior guesses about angles and number of sources generated by inherent limitations of the preprocessing and may lead to catastrophic errors in practical applications.

In this paper, a new approach to wideband direction finding is introduced and described. The proposed strategy combines a robust near-optimal data-adaptive statistic, called the *weighted average of signal subspaces* (WAVES), with an enhanced design of focusing matrices to ensure a statistically robust preprocessing of wideband data. The overall sensitivity of WAVES to various error sources, such as imperfect array focusing, is also reduced with respect to traditional CSSM algorithms, as demonstrated by extensive Monte Carlo simulations.

Index Terms—Array processing, coherent wideband focusing, robust statistics, subspace fitting.

I. INTRODUCTION

THE PROBLEM of locating radiating sources with sensor arrays has been a central topic of signal processing research during the last two decades [1]. Direction-finding algorithms have a straightforward application in sonar and radar [1], [2] and are also useful in advanced satellite and cellular communication systems for the adaptive beamforming of “smart antennas” [3].

Signal and noise subspaces are among the most useful concepts developed for narrowband array processing. They are defined by the eigenvectors of the *spatial data covariance matrix* (SCM) of baseband array outputs [4]. This idea was studied in depth in [5] and [6], giving origin to many relevant algorithms for narrowband direction finding, namely, MUSIC [4], ESPRIT [7], MIN-NORM [8], and WSF [9], that are statistically consistent under the standard assumptions of Gaussian signals and noise and perfect array calibration.

Unlike the maximum likelihood (ML) method [10], which strictly depends on the exact data distribution, subspace algorithms require only the prior knowledge of the array signal model

and the availability of a consistent estimate of the SCM. Furthermore, some subspace algorithms, such as ROOT-MUSIC [11] and ESPRIT, guarantee a fast and global numerical convergence without the need of any prior guesses about source locations.

The subspace approach was subsequently extended to the more difficult wideband environment, where the array manifold changes with frequency and point sources lose their rank-one signature in the SCM [12], [13].

Independent processing of several narrow-frequency bins and clustering of resulting DOA estimates leads to a higher signal-to-noise ratio (SNR) threshold for the resolution of closely spaced sources [14].

Similarly to ML algorithms [15], the wideband generalization of the *weighted subspace fitting* (WSF) criterion [16] requires a cumbersome multidimensional search through the array manifold before starting a local descent algorithm. Since ML and WSF functionals are very flat far from the true DOAs and characterized by many local minima [9], [16], this preprocessing must generate *good* initial guesses for the DOAs (within a *fraction* of the array beam-width of the true values) and an *exact* estimate of the number of sources. Any error at this stage can produce bias, convergence to a local minimum, or a *catastrophe*.

The *coherent signal subspace method* (CSSM), which was proposed by Wang and Kaveh in [14] and further developed in [12], [17], and [18], decomposes wideband array signals in the frequency domain using the DFT or a filterbank so that each subband output satisfies the narrowband model. The SCMs are estimated at several frequency bins, aligned (*focused*) by proper transformation matrices, and finally averaged. This process generates a reduced statistic, referred to as the *universal spatial covariance matrix* (USCM), which (approximately) exhibits the same eigenstructure of the narrowband SCM at a prefixed *focusing frequency*.

The CSSM can exploit the full time-bandwidth product of sources and cope with coherent wavefronts [19]. Moreover, the focusing stage is rather flexible since the reference array geometry may differ from the original one. This property has been exploited for developing computationally efficient algorithms [18], [20].

Unfortunately, the standard CSSM has several drawbacks.

- i) Arrays cannot be generally focused over the entire field of view with small errors by linear transformations [20]. Moreover, statistically optimal focusing matrices are unitary in the isotropic noise case [17], [19]. With this constraint, the approximation error increases because fewer degrees of freedom are available for the design.
- ii) Most signal subspace and ML algorithms are *very sensitive* to model errors [21], [22]. In particular, the number

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of sources may not be consistently detected from the USCM by statistical methods [23] because the background noise becomes “colored” in an unpredictable manner in the presence of focusing errors [24].

- iii) As a consequence, most algorithms focus the array only over (the union of) narrow angular sectors that are selected after a preliminary beamforming search [14]. Unfortunately, beamformer sidelobes can mask weak sources or focusing sectors may be mis-specified, compromising the quality of DOA estimates.
- iv) Source cross-spectra and focusing errors vary across the sensor passband. Strong errors concentrated in only few bins can impair the entire USCM estimate.

From the above discussion, it turns out that the local approximations of the wideband array model around the true DOAs, exploited by ML, WSF, and CSSM, limit the robustness of the overall estimation process.

Resistance to catastrophic errors can be viewed as a problem of reducing the sensitivity of the USCM to a few incorrect SCM estimates. A similar issue has been raised in the framework of *robust statistics* and *cross-validation* of empirical observations [25], [26].

The intrinsic redundancy of wideband array data, due to the large time-bandwidth product of sources, has been exploited in this paper to introduce a new statistic called *weighted average of signal subspaces* (WAVES), which furnishes a basis for the *universal* (focused) signal subspace [19]. Signal subspace eigenvectors estimated from the narrowband decomposition of array outputs and properly focused generate a *pseudodata* matrix, which approximately obeys the narrowband array model. The WAVES estimate is obtained from this matrix through the concept of *pseudocovariance*, which is drawn from robust statistics [25]. In particular, SCM eigenvectors are adaptively weighted to limit the impact of model errors on the final estimate.

The sample WAVES is consistent under the hypothesis of absence of focusing errors when nonunitary focusing matrices are used. Moreover, it is less sensitive than the USCM against actual focusing errors, mistakes in the selection of signal subspace eigenvectors, and the presence of weak narrowband sources.

The high performance of WAVES is demonstrated by theoretical considerations and confirmed by extensive computer simulations.

Another critical issue for the statistical performance of coherent wideband processing is the synthesis of focusing matrices. Classical least squares (LS) fitting [17], [18], [20], [27] is not directly related to the optimization of statistical performance. Therefore, a new angle-dependent error criterion for coherent focusing is presented. It relies on standard results of the asymptotical analysis of subspace algorithms [9], [11] to reduce the bias of *beamforming invariance* CSSM focusing [18], which does not require a preliminary search over large angular sectors.

II. BACKGROUND

A. Notation

Throughout the present work, matrices will be indicated by capital boldface letters and vectors by lower-case boldface letters. Other symbols are as follows.

$E[\cdot]$ Statistical expectation operator.

- $(\cdot)^H$ Hermitian matrix transpose.
 - $(\cdot)^\dagger$ Moore–Penrose pseudoinverse of a matrix.
 - $\mathbf{I}_m$ $m \times m$ identity matrix.
 - $\|\cdot\|_F$ Frobenius norm of a matrix [29].
 - δ_{ij} Kronecker symbol.
- Sample quantities will be marked by a tilde superscript.

B. Signal Model

An M -element array receives D signals, radiated by point sources. Each sensor signal is converted to the baseband, is time sampled, and is frequency decomposed, using a filterbank or the DFT [2], into J complex subband components, sampled with period T . The bandwidth of each filter is much smaller than its central frequency f_i ($i = 1, 2, \dots, J$) so that each subband *snapshot* $\mathbf{x}_i(nT)$ ($n = 0, 1, \dots, N-1$) approximately satisfies the classical narrowband equation [4]

$$\mathbf{x}_i(nT) = \mathbf{A}(f_i)\mathbf{s}_i(nT) + \mathbf{n}_i(nT) \quad (1)$$

where

$$\mathbf{A}(f_i) = [\mathbf{a}_1(f_i, \theta_1), \dots, \mathbf{a}_D(f_i, \theta_D)] \quad (2)$$

is the $M \times D$ array transfer matrix at frequency f_i , and $\mathbf{s}_i(nT)$ is the vector of complex source signals. Columns of $\mathbf{A}(f_i)$ are referred to as *steering vectors* and represent the array response to each wavefront. It is assumed that the generic steering vector $\mathbf{a}(f, \theta)$ has unit L_2 norm and is a *known*, continuous, and differentiable function of the frequency f and the P location parameters $\{\phi_p; p = 1, 2, \dots, P\}$ contained in vector θ (azimuth, elevation, range, ...). For unique identifiability of the model, $\mathbf{A}(f_i)$ must have full rank for any set of $D+1 \leq M$ source locations [4]. Finally, $\mathbf{n}_i(nT)$ is the $M \times 1$ vector of additive noise at frequency f_i and time nT .

Sources and noise are assumed to be realizations of independent, zero-mean, complex, circular random processes having at least finite second- and fourth-order moments [6]. The background noise is considered white and isotropic at any frequency, with variance σ_n^2 [4]. Colored noise, whose spatio-temporal covariance is known, can be handled by prewhitening [4], [6].

If the subband sampling period T is *much* larger than the correlation time of sources and filter passbands do not overlap, snapshots $\mathbf{x}_i(nT)$ can be considered independent with respect to both time and frequency indexes.

Under these standard hypotheses, the SCM at frequency f_i , $\mathbf{R}_{xx}(f_i)$ obeys the equation

$$\begin{aligned} \mathbf{R}_{xx}(f_i) &= E[\mathbf{x}_i(nT)\mathbf{x}_i^H(nT)] \\ &= \mathbf{A}(f_i)\mathbf{R}_{ss}(f_i)\mathbf{A}^H(f_i) + \sigma_n^2\mathbf{I}_M. \end{aligned} \quad (3)$$

The eigenvectors corresponding to the $\eta_i \leq D$ largest SCM eigenvalues $\lambda_1(f_i) \geq \lambda_2(f_i) \geq \dots \geq \lambda_{\eta_i}(f_i)$, collected in the matrix $\mathbf{E}_s(f_i)$, constitute a basis for the *signal subspace* [4] at frequency f_i . The signal subspace is contained in the range of $\mathbf{A}(f_i)$. If wavefronts are not completely spatially coherent, the source covariance $\mathbf{R}_{ss}(f_i)$ has rank D , and the range of $\mathbf{A}(f_i)$ coincides with the signal subspace [4].

The orthogonal complement to the signal subspace is called the *noise subspace* and is defined by the basis $\mathbf{E}_n(f_i)$ of the $M - \eta_i$ eigenvectors that are associated with the smallest eigenvalue σ_n^2 of $\mathbf{R}_{xx}(f_i)$.

In practice, $\mathbf{R}_{xx}(f_i)$ is estimated from $N \gg M$ snapshots. For the validity of the following discussion, the SCM estimate $\hat{\mathbf{R}}_{xx}(f_i)$ must be consistent and asymptotically unbiased.

C. CSSM

The CSSM [14] uses $M \times M$ transformation matrices $\mathbf{T}(f_i)$ to map each steering vector $\mathbf{a}(f_i, \theta)$ to the corresponding steering vector $\mathbf{b}(f_0, \theta)$ of a *reference* array, operating at the focusing frequency f_0 . Using (2), *perfect focusing* is attained when for each f_i

$$\begin{aligned} \mathbf{B}(f_0) &= \mathbf{T}(f_i)\mathbf{A}(f_i) \\ &= [\mathbf{b}(f_0, \theta_1) \cdots \mathbf{b}(f_0, \theta_D)]. \end{aligned} \quad (4)$$

Ideally, all transformed snapshots $\mathbf{T}(f_i)\mathbf{x}_i(nT)$ obey the same model (1). However, (4) cannot be exactly satisfied for geometrical reasons [27], and an irreducible *focusing error* is introduced.

The USCM, which is indicated by $\mathbf{R}_F$, is the weighted average of focused $\mathbf{R}_{xx}(f_i)$ s with preselected scalar positive weights α_i s

$$\begin{aligned} \mathbf{R}_F &= \sum_{i=1}^J \alpha_i \mathbf{T}(f_i) \mathbf{R}_{xx}(f_i) \mathbf{T}^H(f_i) \\ &\cong \mathbf{B}(f_0) \left(\sum_{i=1}^J \alpha_i \mathbf{R}_{ss}(f_i) \right) \mathbf{B}^H(f_0) + \sigma_n^2 \mathbf{R}_N. \end{aligned} \quad (5)$$

If the $M \times M$ matrix $\mathbf{Q}$ is a *square root* [26] of the universal noise spatial covariance $\mathbf{R}_N$, according to [18], [20]

$$\mathbf{R}_N = \mathbf{Q}\mathbf{Q}^H = \sum_{i=1}^J \alpha_i \mathbf{T}(f_i) \mathbf{T}(f_i)^H \quad (6)$$

the *universal signal subspace* $\mathbf{E}_S$ and its orthogonal complement $\mathbf{E}_N$ (*universal noise subspace*) are obtained from the eigendecomposition of the matrix $\mathbf{Q}^{-1}\mathbf{R}_F\mathbf{Q}^{-H}$.

The sample versions of $\mathbf{E}_S$ and $\mathbf{E}_N$, respectively, $\tilde{\mathbf{E}}_S$ and $\tilde{\mathbf{E}}_N$, can be used in any subspace algorithm for direction finding. For instance, the MUSIC estimate becomes

$$\tilde{\theta}_{\text{MUSIC}} = \arg \min_{\theta} \left\{ \frac{|\tilde{\mathbf{E}}_N^H \mathbf{Q}^{-1} \mathbf{b}(f_0, \theta)|_2}{|\mathbf{Q}^{-1} \mathbf{b}(f_0, \theta)|_2} \right\}. \quad (7)$$

In [18], an enhanced focusing scheme called *beamforming invariance* CSSM (BI-CSSM) was introduced. In this approach, proper $K \times M$ beamforming matrices $\mathbf{T}(f_i)$ and $\mathbf{T}_0$ are synthesized offline and applied to both actual and reference steering vectors so that

$$\mathbf{T}(f_i)\mathbf{a}(f_i, \theta) \cong \mathbf{T}_0\mathbf{b}(f_0, \theta). \quad (8)$$

It must be $M \geq K > D$ for model identifiability. This “balanced” transformation enhances the flexibility of the design and reduces focusing errors over large angular sectors. Therefore, the preliminary beamforming can often be avoided.

In [24], a *covariance matching* approach for building unitary focusing matrices has been proposed. This solution uses only estimated narrowband signal subspaces and enhances the performance at high SNR. In particular, it furnishes low-bias DOA estimates. Nevertheless, estimation errors can impair the performance with short data records.

D. Wideband WSF Criterion

The classical CSSM spatially transforms both signals and noise. In contrast, the narrowband WSF algorithm [9] uses *only signal subspace eigenvectors* to get asymptotically efficient DOA estimates $\hat{\Theta}_{\text{WSF}}$. Extension of the WSF criterion to the wideband case is straightforward [16] since the global and concentrated Gaussian log-likelihood is simply the sum of log-likelihoods of narrowband components if snapshots can be considered independent among different frequency bins [19]:

$$\hat{\Theta}_{\text{WSF}} = \arg \min_{\Theta} \left\{ \sum_{i=1}^J \left| \mathbf{A}(f_i, \Theta) \tilde{\mathbf{C}}_i - \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i) \right|_F^2 \right\}. \quad (9)$$

$\tilde{\mathbf{E}}_s(f_i)$ is the estimated signal subspace at frequency f_i . For Gaussian signals and noise and perfectly calibrated array, optimal $\eta_i \times \eta_i$ weighting matrices $\mathbf{P}(f_i)$ are diagonal with elements proportional to [9]

$$\mathbf{P}(f_i)_{[k,k]} = \frac{\lambda_k(f_i) - \sigma_n^2}{\sqrt{\lambda_k(f_i) \sigma_n^2}}. \quad (10)$$

Each $D \times \eta_i$ matrix $\tilde{\mathbf{C}}_i$ in (9) satisfies the relationship $\tilde{\mathbf{C}}_i = \mathbf{A}^\dagger(f_i, \Theta) \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i)$. Optimality is preserved even if consistent estimates of weights are substituted in place of true values [9].

III. WEIGHTED AVERAGE OF SIGNAL SUBSPACES

The diagonal shape of the optimal $\mathbf{P}(f_i)$ in (10) means that sample signal subspace eigenvectors work like *independent observations* [1], at least for Gaussian signals.¹ This property can be exploited to develop a new global statistic, which can effectively replace the USCM.

Solution of (9) does not change if single terms of the summation are premultiplied by unitary $M \times M$ matrices $\mathbf{T}(f_i)$ that satisfy the perfect focusing condition (4).² Equation (9) becomes

$$\begin{aligned} \hat{\Theta}_{\text{WSF}} &= \arg \min_{\Theta} \left\{ \sum_{i=1}^J \left| \mathbf{B}(f_0, \Theta) \tilde{\mathbf{C}}_i - \mathbf{T}(f_i) \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i) \right|_F^2 \right\}. \end{aligned} \quad (11)$$

¹This property stems from the fact that the particular Gaussian ML estimate of the SCM is invariant under unitary similarity transforms of data [28], which is not true in general [25].

²More general transformations are also possible, but they lead to unnecessary mathematical complications.

Formula (11) suggests the introduction of the matrix

$$\begin{aligned}\tilde{\mathbf{Z}} &\triangleq \left(\sum_{i=1}^J \eta_i \right)^{-1/2} \\ &\cdot \left[\mathbf{T}(f_1) \tilde{\mathbf{E}}_s(f_1) \mathbf{P}(f_1), \dots, \mathbf{T}(f_J) \tilde{\mathbf{E}}_s(f_J) \mathbf{P}(f_J) \right] \\ &= \eta^{-1/2} [\tilde{\mathbf{Z}}_1, \dots, \tilde{\mathbf{Z}}_J].\end{aligned}\quad (12)$$

The next two theorems state that the $d \leq D$ principal left singular vectors [26] of $\tilde{\mathbf{Z}}$ asymptotically furnish an estimate of the *universal signal subspace*.

Theorem 1: The sample matrix $\tilde{\mathbf{Z}}$ converges with probability 1 (w.p. 1) to a *fixed* matrix $\mathbf{Z}$ under the following assumptions.

- 1) Each sample subspace $\tilde{\mathbf{E}}_s(f_i)$ converges w.p. 1 to the true $\mathbf{E}_s(f_i)$ as $N \rightarrow \infty$.
- 2) Matrices $\mathbf{T}(f_i)$ and $\mathbf{P}(f_i)$ have full rank and finite L_2 norm.
- 3) SCM eigenvectors are made unique by a proper scaling [28].

Proof: From Assumption 1, $|\tilde{\mathbf{E}}_s(f_i) - \mathbf{E}_s(f_i)|_2 \rightarrow 0$ w.p. 1, as $N \rightarrow \infty$ [10]. Therefore, each $M \times \eta_i$ block $\tilde{\mathbf{Z}}_i = \mathbf{T}(f_i) \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i)$ tends, in the same limit, to the matrix $\mathbf{Z}_i \triangleq \mathbf{T}(f_i) \mathbf{E}_s(f_i) \mathbf{P}(f_i)$ since

$$\begin{aligned}|\tilde{\mathbf{Z}}_i - \mathbf{Z}_i|_2 &\leq |\mathbf{T}(f_i)|_2 |\mathbf{P}(f_i)|_2 |\tilde{\mathbf{E}}_s(f_i) - \mathbf{E}_s(f_i)|_2 \\ &\leq \text{const} \times |\tilde{\mathbf{E}}_s(f_i) - \mathbf{E}_s(f_i)|_2 \rightarrow 0 \quad \text{w.p. 1.}\end{aligned}$$

Therefore, the entire $\tilde{\mathbf{Z}}$ stochastically converges to the unique $M \times \eta$ matrix $\mathbf{Z} = \eta^{-1/2} [\mathbf{Z}_1 \dots \mathbf{Z}_J]$ (see Appendix A for a formal proof). ■

Theorem 2: If, in addition to assumptions of *Theorem 1*, focusing matrices $\mathbf{T}(f_i)$ satisfy (4), all columns of $\mathbf{Z}$ lie in the range of $\mathbf{B}(f_0)$.

Proof: Since each term of the summation in (9) converges to 0 w.p. 1 [9], it follows that [26]

$$\begin{aligned}|\mathbf{A}(f_i, \Theta) \tilde{\mathbf{C}}_i - \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i)|_2 \\ \leq |\mathbf{A}(f_i, \Theta) \tilde{\mathbf{C}}_i - \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i)|_F \rightarrow 0 \quad \text{w.p. 1.}\end{aligned}$$

Since $\mathbf{T}(f_i)$ has full rank and bounded L_2 norm, the following inequalities hold w.p. 1, after inserting (4):

$$\begin{aligned}|\mathbf{B}(f_0) \tilde{\mathbf{C}}_i - \tilde{\mathbf{Z}}_i|_2 &= |\mathbf{T}(f_i) \mathbf{A}(f_i, \Theta) \tilde{\mathbf{C}}_i - \mathbf{T}(f_i) \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i)|_2 \\ &\leq |\mathbf{T}(f_i)|_2 |\mathbf{A}(f_i, \Theta) \tilde{\mathbf{C}}_i - \tilde{\mathbf{E}}_s(f_i) \mathbf{P}(f_i)|_2 \rightarrow 0 \\ |\mathbf{B}(f_0) \tilde{\mathbf{C}}_i - \mathbf{Z}_i|_2 &\leq |\mathbf{B}(f_0) \tilde{\mathbf{C}}_i - \tilde{\mathbf{Z}}_i|_2 + |\tilde{\mathbf{Z}}_i - \mathbf{Z}_i|_2 \rightarrow 0 \\ |\tilde{\mathbf{C}}_i - \mathbf{C}_i|_2 &\leq |\mathbf{A}^\dagger(f_i, \Theta)|_2 |\tilde{\mathbf{E}}_s(f_i) - \mathbf{E}_s(f_i)|_2 |\mathbf{P}(f_i)|_2 \rightarrow 0.\end{aligned}$$

The only tenable conclusion is that

$$\mathbf{Z} = \eta^{-1/2} \mathbf{B}(f_0) [\mathbf{C}_1 \dots \mathbf{C}_J] = \mathbf{B}(f_0) \mathbf{C}.$$

■

Theorem 2 indicates that the d -dimensional subspace spanned by columns of $\mathbf{Z}$ defines a basis for the universal signal subspace. Since this subspace can be interpreted as the average of weighted $\mathbf{E}_s(f_i)$ s, from now on, it will be referred to as the WAVES.

IV. DIRECTION FINDING USING WAVES

The sample matrix $\tilde{\mathbf{Z}}$ is always full rank because of the presence of finite sample and focusing errors. In particular, $\tilde{\mathbf{Z}}$ can be considered to be a *pseudodata* matrix, built with sample SCM eigenvectors, instead of raw array snapshots. Therefore, from *Theorems 1* and *2*, it is quite natural to estimate a basis for the WAVES from the d principal left singular vectors $\tilde{\mathbf{U}}_S$ of the following *reduced-size* SVD (R-SVD) of $\tilde{\mathbf{Z}}$ [26]

$$\tilde{\mathbf{Z}} = [\tilde{\mathbf{U}}_S \tilde{\mathbf{U}}_N] \begin{bmatrix} \tilde{\Sigma}_S & \mathbf{0} \\ \mathbf{0} & \tilde{\Sigma}_N \end{bmatrix} [\tilde{\mathbf{W}}_S \tilde{\mathbf{W}}_N]^H. \quad (13)$$

This algorithm can be regarded as a *total least squares* (TLS) approximation step [7], [26], attempting to “restore” the original rank- d property of $\mathbf{Z}$. It will be shown in Section V that $\tilde{\mathbf{U}}_S$ asymptotically converges to the WAVES.

The basis $\tilde{\mathbf{U}}_S$ and its orthogonal complement $\tilde{\mathbf{U}}_N$ can replace the corresponding CSSM universal subspaces $\tilde{\mathbf{E}}_S$ and $\tilde{\mathbf{E}}_N$ in any subspace-based algorithm for direction finding.

For instance, a very basic algorithm combining WAVES and MUSIC can be summarized as follows.

Step 1) For $i = 1, 2, \dots, J$, estimate $\tilde{\mathbf{E}}_s(f_i)$ and η_i from the eigendecomposition (EVD) of the sample SCM

$$\tilde{\mathbf{R}}_{xx}(f_i) = \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{x}_i(nT) \mathbf{x}_i(nT)^H.$$

Step 2) For $i = 1, 2, \dots, J$, compute $\mathbf{T}(f_i)$ and $\mathbf{P}(f_i)$ according to some available criterion.

Step 3) Form the matrix $\tilde{\mathbf{Z}}$ according to (12).

Step 4) Estimate the number of sources $\hat{D}$, the WAVES $\tilde{\mathbf{U}}_S$, and $\tilde{\mathbf{U}}_N$ from the R-SVD of $\tilde{\mathbf{Z}}$ (13).

Step 5) Estimate angles using the MUSIC criterion (7) with $\tilde{\mathbf{U}}_N$ in place of $\tilde{\mathbf{E}}_N$.

Remarks:

- 1) An important issue for the consistency of DOA estimators based on $\tilde{\mathbf{Z}}$ is that the number of free parameters of matrix $\mathbf{C}$ remains finite, regardless of the number of snapshots used to form each SCM.
- 2) *Theorems 1* and *2* can be generalized to other focusing schemes even if focusing matrices are chosen nonunitary and/or nonsquare [18], [20], provided that the rank of $\mathbf{T}(f_i)$ is greater than D and all matrices $\mathbf{T}(f_i) \mathbf{A}(f_i)$ have rank D (i.e., none of steering vectors has been canceled out by the transformation).
- 3) *Theorem 1* also shows that $\tilde{\mathbf{Z}}$ converges regardless of the noise covariance, since biasing effects have been removed during the EVD of $\tilde{\mathbf{R}}_{xx}(f_i)$. It is stressed that this convergence property is not shared by the classical USCM and

avoids the need for a generalized SVD [6], [26] in most circumstances. Clearly, the optimal subspace weighting for the final DOA estimation remains an open issue.

- 4) By the same arguments used in [19], the rank d of $\mathbf{Z}$ can reach D , even in the presence of spatially coherent sources, if $\eta \geq D$. In this case, the WAVES spans exactly the range of $\mathbf{B}(f_0)$. However, the actual capability of decorrelating coherent sources also depends on the frequencies selected for focusing. If they are too close and/or source spectra are nearly proportional, the smoothing may not be effective.
- 5) Focusing matrices can be synthesized as described in [17] and [20] after a preliminary beamforming. However, well-designed beam-space transformations [18] can be implemented by noniterative algorithms, thus reducing on-line computations.
- 6) Fast iterative algorithms to compute the EVD of $\tilde{\mathbf{R}}_{xx}(f_i)$ and the R-SVD (13) are described in [26], [29], and [30].

V. ERROR ANALYSIS OF WAVES

A rigorous error analysis of WAVES appears mathematically intractable. However, an approach based on standard first-order perturbative expansions [11], [26] of sample signal subspaces and WAVES worked satisfactorily under the hypothesis of *sufficiently small* estimation errors. This analysis shed light on the asymptotical performance of WAVES and allowed the synthesis of algorithms resistant also to large errors.

The analysis is divided into three parts:

- 1) sensitivity of sample signal subspaces;
- 2) TLS estimation of WAVES;
- 3) asymptotical properties of DOA estimates.

A. Sample Subspace Errors

The sample signal subspace $\tilde{\mathbf{E}}_s(f_i)$ is expanded in Taylor series with respect to a *dummy variable* [26]. Since first-order perturbations generated by independent mechanisms can be linearly superimposed on $\tilde{\mathbf{E}}_s(f_i)$, it is possible to consider main error sources separately, starting from the array model (1).

1) *Finite Sample Errors*: The Taylor series of $\tilde{\mathbf{E}}_s(f_i)$ can be written as

$$\tilde{\mathbf{E}}_s(f_i) = \mathbf{E}_s(f_i) + \varepsilon [\mathbf{E}_s(f_i) \quad \mathbf{E}_n(f_i)] \mathbf{G}(f_i) + \dots \quad (14)$$

The dummy variable $\varepsilon \ll 1$ can be conveniently related to the number of snapshots as $\varepsilon = 1/\sqrt{N}$. If $N \gg M$, it is possible to retain only first-order terms.

First- and second-order statistics of the *random* $M \times \eta_i$ matrix $\mathbf{G}(f_i)$ are needed for performance analysis [11]. In particular, if $\lambda_k(f_i)$ and $\lambda_l(f_i)$ ($k, l \leq \eta_i$) are two *distinct* signal eigenvalues at frequency f_i and $m, n > \eta_i$, the following statistics hold in the Gaussian case [11], [28]:

$$E \{ \mathbf{G}(f_i)_{[n,k]} \} = 0 \quad (15)$$

$$E \{ \mathbf{G}(f_i)_{[n,k]} \mathbf{G}(f_i)_{[m,l]}^* \} = 0 \quad (16)$$

$$E \{ \mathbf{G}(f_i)_{[n,k]} \mathbf{G}(f_i)_{[m,l]}^* \} = \delta_{nm} \delta_{kl} \frac{\sigma_n^2 \lambda_k(f_i)}{[\lambda_k(f_i) - \sigma_n^2]^2}. \quad (17)$$

Equation (15) means that sample signal subspaces are asymptotically unbiased. Equations (16) and (17) indicate that spurious projections from distinct signal eigenvectors onto $\mathbf{E}_n(f_i)$ can be asymptotically modeled as *mutually uncorrelated* random vectors [9]. Components of these vectors are also asymptotically circular and mutually uncorrelated and have the *same variance* if weighting (10) and unitary focusing matrices are adopted for building $\tilde{\mathbf{Z}}$.³

2) *Calibration and Focusing Errors*: Finite sample errors are now neglected (e.g., $\varepsilon \rightarrow 0$). Each summation term in (9) is therefore zero:

$$\mathbf{A}(f_i) \mathbf{C}_i = \mathbf{E}_s(f_i) \mathbf{P}(f_i). \quad (18)$$

According to (4) and (11), small focusing (or calibration) errors can be modeled as [31]

$$\mathbf{T}(f_i) \mathbf{A}(f_i) - \mathbf{B}(f_0) = \Delta \mathbf{B}(f_i) = \mu \mathbf{B}^{(1)}(f_i). \quad (19)$$

The scale factor μ is chosen so that

$$\mu \|\mathbf{B}(f_0)\|_2 = \max_{1 \leq i \leq J} \|\Delta \mathbf{B}(f_i)\|_2$$

and typically lies in the range $(10^{-1} \div 10^{-3})$ within the focusing sector(s) [18], [20].

Since $\mathbf{A}(f_i)$ is full rank by hypothesis, its pseudoinverse $\mathbf{A}^\dagger(f_i)$ has finite L_2 norm [26]. After combining (11), (18), and (19), the perturbation of each block $\mathbf{Z}_i$ can be expressed as

$$\tilde{\mathbf{Z}}_i - \mathbf{Z}_i = \mu \mathbf{B}^{(1)}(f_i) \mathbf{A}^\dagger(f_i) \mathbf{E}_s(f_i) \mathbf{P}(f_i). \quad (20)$$

Equation (20) shows that the perturbation critically depends not only on the focusing error itself but on eigenvector weights and the L_2 norm of $\mathbf{A}^\dagger(f_i)$ as well, which generally increases in the presence of closely spaced sources and at the low-end of the sensor passband.

This model can be applied also to the WSF functional (9). Since, in general, errors of different signal eigenvectors are correlated, optimal weighting matrices are no longer diagonal, as in (10) [22]. DOA estimates obtained by WSF also become biased because the mean of fitting residuals is generally nonzero due to the likely presence of systematic calibration errors.

3) *Other Error Sources*: Finite bandwidth effects within a single bin, aliasing, and spectral leakage from filter or window sidelobes may affect the narrowband SCM estimate. Proper perturbative expansions are not reported here because of space limitations. However, final effects on $\tilde{\mathbf{E}}_s(f_i)$ are qualitatively similar to those induced by focusing errors.

B. TLS Estimation of WAVES

The combination of (14) and (20) gives

$$\begin{aligned} \tilde{\mathbf{Z}}_i &= \mathbf{Z}_i(\varepsilon, \mu) \\ &= \mathbf{Z}_i + \left\{ \varepsilon \mathbf{T}(f_i) [\mathbf{E}_s(f_i) \quad \mathbf{E}_n(f_i)] \mathbf{G}(f_i) \right. \\ &\quad \left. + \mu \mathbf{B}^{(1)}(f_i) \mathbf{A}^\dagger(f_i) \mathbf{E}_s(f_i) \right\} \mathbf{P}(f_i) + \dots \end{aligned} \quad (21)$$

³Over different frequency bins, subspace fitting errors are independent, and the noise power is constant for hypotheses made in Section II. Unitary focusing matrices preserve orthogonality between signal and noise subspaces [6], [17].

Signal subspace errors generate spurious components in the original nullspace of $\mathbf{Z}$ defined by the $(M - d)$ -dimensional basis $\mathbf{U}_N$ and are referred to from now on as the *error subspace*.

Substitution of (21) into (12) shows that $\tilde{\mathbf{Z}}$ can be expanded in Taylor series with respect to the same variables ε and μ defined in Section V-A. In particular, the R-SVD (13) can be rewritten with the notation used in (21) as

$$\mathbf{Z}(\varepsilon, \mu) \mathbf{W}(\varepsilon, \mu) = \mathbf{U}(\varepsilon, \mu) \mathbf{\Sigma}(\varepsilon, \mu) \quad (22)$$

where

$$\mathbf{Z}(0, 0) \equiv \mathbf{Z} = [\mathbf{U}_S \ \mathbf{U}_N] \begin{bmatrix} \mathbf{\Sigma}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma}_N \end{bmatrix} [\mathbf{W}_S \ \mathbf{W}_N]^H. \quad (23)$$

The analysis is carried out as in Section V-A. In particular, the sample WAVES $\tilde{\mathbf{U}}_S$ can be related to the true $\mathbf{U}_S$ and $\mathbf{U}_N$ by defining the $M \times d$ error matrix $\mathbf{H}^{(1)}(\varepsilon, \mu) = \varepsilon \mathbf{H}_\varepsilon^{(1)} + \mu \mathbf{H}_\mu^{(1)}$ through the equation

$$\tilde{\mathbf{U}}_S = \mathbf{U}_S + [\mathbf{U}_S \ \mathbf{U}_N] \mathbf{H}^{(1)}(\varepsilon, \mu) + \dots \quad (24)$$

Some statistics of elements of $\mathbf{H}^{(1)}(\varepsilon, \mu)$ are required for performance analysis [9], [11]. An in-depth study of (22) and (23) reveals interesting properties of the sample WAVES and sheds light on some identifiability issues. In particular, we have the following.

- 1) The small-error assumption breaks down if $|\mathbf{H}^{(1)}(\varepsilon, \mu)|_\infty$ approaches unity. Considering unit-norm eigenvectors, standard bounds (see [26] and Appendix A) lead to the condition

$$|\mathbf{H}^{(1)}(0, \mu)|_{[j, i]} \leq \sqrt{\frac{J}{\eta}} \cdot \max_{1 \leq i \leq J} \frac{\mu |\mathbf{T}(f_i)|_2 |\mathbf{P}(f_i)|_2}{\sigma_d} \ll 1 \quad (25)$$

for $d < j \leq M$. Bound (25) can be violated if the L_2 norms of $\mathbf{T}(f_i)$ and/or $\mathbf{P}(f_i)$ are large, as in some suboptimal wideband focusing schemes [17], [20] and high SNR environments. It is also shown by (21) that frequency-dependent focusing errors usually affect only a few columns of $\tilde{\mathbf{Z}}$.⁴

- 2) Successful estimation of the number of sources D from the R-SVD of $\tilde{\mathbf{Z}}$ requires that the D th unperturbed singular value σ_D of $\mathbf{Z}$ ($\sigma_1 \geq \dots \geq \sigma_D \geq \dots \geq \sigma_M$) fairly exceeds the $(D + 1)$ th sample singular value $\tilde{\sigma}_{D+1}$ of $\tilde{\mathbf{Z}}$ [6]. From well-known SVD properties [26] and the model (22), it must be

$$\tilde{\sigma}_{D+1} \cong \left| \mathbf{U}_N^H (\tilde{\mathbf{Z}} - \mathbf{Z}) \mathbf{W}_N \right|_2 \ll \sigma_D. \quad (26)$$

- 3) Under the Gaussian stochastic assumption for signals and noise [10], the asymptotical analysis of WAVES can be easily carried out, if weighting (10) is used, and unitary focusing matrices satisfy conditions listed in [19]. In particular, the assumption of *perfect subspace mapping* [19] implies that $\mu = 0$ in (24).

In this case, asymptotically, it is $E[\tilde{\sigma}_n] = 1/\sqrt{N}$ for $d < n \leq M$. Therefore, the sample error subspace of $\tilde{\mathbf{Z}}$ asymptotically behaves as the noise subspace of the narrowband SCM. According to (26), this property holds when $\sigma_d \gg 1/\sqrt{N}$ and

⁴Errors typically grow with $|f_i - f_0|$ and at the boundaries of focusing sectors.

might be exploited for the detection of the number of sources. The fundamental statistics of the projections of $\tilde{\mathbf{U}}_S$ onto the true error subspace basis $\mathbf{U}_N$ can be easily worked out as in [28], obtaining

$$E \left\{ \mathbf{H}_{[n, k]}^{(1)}(\varepsilon, 0) \right\} = 0 \quad (27)$$

$$E \left\{ \mathbf{H}_{[n, k]}^{(1)}(\varepsilon, 0) \mathbf{H}_{[m, l]}^{(1)}(\varepsilon, 0) \right\} = 0 \quad (28)$$

$$E \left\{ \mathbf{H}_{[n, k]}^{(1)}(\varepsilon, 0) \mathbf{H}_{[m, l]}^{(1)*}(\varepsilon, 0) \right\} = \delta_{nm} \delta_{kl} \frac{1}{\eta N \sigma_k^2} \quad (29)$$

for $1 \leq k, l \leq d$ and $n, m > d$.

C. Asymptotical Performance

Under the same assumptions made to derive (27)–(29), asymptotical properties of DOA estimators can be found by the methods reported in [9] and [11]. For instance, the analysis of WSF using WAVES (WAVES-WSF) is easily carried out from the Taylor expansion around the true angles of the estimator

$$\tilde{\Theta}_{\text{WAVES-WSF}} = \arg \min_{\Theta} \left\{ \left| \mathbf{B}(f_0, \Theta) \tilde{\mathbf{C}}_W - \tilde{\mathbf{U}}_S \mathbf{P}_W \right|_F^2 \right\}. \quad (30)$$

Equation (27) indicates that WAVES-WSF is asymptotically unbiased [28]. From (28) and (29), the optimal $d \times d$ weighting matrix turns out to be $\mathbf{P}_W = \mathbf{\Sigma}_S$. Therefore, the algorithm is asymptotically equivalent to a narrowband WSF using $N\eta$ independent snapshots. Asymptotical performance is not affected by the choice $\mathbf{P}_W = \tilde{\mathbf{\Sigma}}_S$ but slightly depends on the particular selection of $\mathbf{T}(f_i)$ s through the $D \times d$ matrix

$$\mathbf{C}_W = E[\tilde{\mathbf{C}}_W] = \mathbf{B}^\dagger(f_0) \mathbf{U}_S \mathbf{P}_W.$$

Numerical evaluation is in close agreement with simulations above the SNR threshold [31] and demonstrates that asymptotical DOA standard deviation is slightly worse ($5 \div 20\%$) than the Cramér–Rao bound (CRB).

Comparing (14) and (20) for typical values of N , source powers, and μ , it is easily realized that calibration and focusing errors generally dominate. In this case, *Theorem 2* loses most of its validity. The main issue is that weighting matrices $\mathbf{P}(f_i)$ s can catastrophically amplify focusing errors in $\tilde{\mathbf{Z}}$. In particular, weighting (10) emphasizes high SNR components and may destroy the sphericity of the sample error subspace, as shown by (26). This effect could be compensated by reducing the weight of corrupted SCM eigenvectors in $\tilde{\mathbf{Z}}$ [25]. Anyway, this procedure may adversely affect the asymptotical performance by reducing the “effective” SNR of some sources, as revealed by $\mathbf{\Sigma}_S$ and (29).

At low SNR, instead, finite sample errors generally prevail, and weighting (10) generates a nearly spherical error subspace in $\tilde{\mathbf{Z}}$.

These properties were well-confirmed by simulations and justified the development of an automatic procedure for weight optimization.

VI. ROBUST ESTIMATION OF WAVES

Signal subspace misalignment essentially produces a leakage in the error subspace, which can mask weak signal components

and generate “ghost” sources. These effects resemble those induced by the presence of *outliers* in multichannel data matrices [25], [32].

Unfortunately, most of the theory developed for robust covariance estimation cannot be directly extended to the problem at hand since fundamental probabilistic concepts, like “distribution,” “contamination,” and “robustness” itself become intrinsically vague when applied to the matrix $\tilde{\mathbf{Z}}$. In particular, a “robust” WAVES estimator should combat mismodeling, starting from a *good estimate* of narrowband eigenvectors (possibly ML). It is stressed that non-Gaussian data and the presence of *outliers* in received signals affect, instead, narrowband SCM estimates and require different approaches [25], [32].

Nevertheless, mathematical tools developed in robust statistics can be *extremely effective* as heuristic optimization procedures if linked to the WAVES concept.

From the previous analysis, it is likely that some corrupted eigenvectors exist and “blow up” the WAVES estimate obtained by the SVD. However, since transformed eigenvectors can be still considered to be *nearly independent observations* of the array response in frequency and space, their impact on the WAVES can be effectively controlled by adaptively scaling each column of $\tilde{\mathbf{Z}}$. This operation is equivalent to optimizing diagonal matrices $\mathbf{P}(f_i)$ and ensures that the WAVES estimate is invariant with respect to unitary column transformations of $\tilde{\mathbf{Z}}$, just as the sample Gaussian ML SCM and the USCM [25], [32]. If most eigenvector weights remain nonzero, *Theorem 2* still guarantees the asymptotical convergence of WAVES.

According to above considerations, the following procedure was developed.

A. Algorithm Initialization

Matrix $\tilde{\mathbf{Z}}$ is built according to (12). From (21)–(29), it makes sense to define a proper *reference model* for the empirical WAVES, assuming Gaussian signals and the absence of focusing errors. Therefore, weighting matrices can be initialized, according to (10). As a matter of fact, this initialization is not critical since the subsequent optimization process demonstrated a very wide *trust region* [33].

B. Weight Optimization

The proposed weight optimization is carried out by solving the following set of implicit equations for the $M \times M$ matrix $\tilde{\mathbf{X}}$:

$$\begin{cases} \tilde{\mathbf{X}} \tilde{\mathbf{y}}_k = \mathbf{T}_k \tilde{\mathbf{e}}_k \rho_k \\ \sum_{k=1}^{\eta} s(|\tilde{\mathbf{y}}_k|)^2 \tilde{\mathbf{y}}_k \tilde{\mathbf{y}}_k^H = \sum_{k=1}^{\eta} s(|\tilde{\mathbf{y}}_k|)^2 \mathbf{I}_M \end{cases} \quad (31)$$

where $\tilde{\mathbf{e}}_k$ is the generic unit-norm sample eigenvector, which, focused by the proper matrix $\mathbf{T}_k$ and scaled by the initial weight ρ_k , constitutes the k th column of $\tilde{\mathbf{Z}}$ ($1 \leq k \leq \eta$). $s(x) \geq 0$ is a proper real-valued scalar function described in the following.

From (31), matrix $\tilde{\mathbf{X}}$ exhibits the same SVD structure of $\tilde{\mathbf{Z}}$. By *Theorem 1*, the proper WAVES estimate is defined by the d principal left singular vectors of $\tilde{\mathbf{X}}$.

Existence and local uniqueness of the solution of (31) have been studied in [25, ch. 8] for the robust M estimation (ML type) of the *pseudocovariance* of elliptical distributions. In the array processing context, system (31) corresponds to an adaptive *cross-validation* [26] in frequency and space of focused narrowband signal subspaces.

A modern and effective *square root* algorithm [26] for solving (31), based on an iteratively reweighted R-SVD of $\tilde{\mathbf{Z}}$, is briefly described in Appendix B.

C. Choice of $s(x)$

The scalar function $s(x)$ should satisfy the convergence requirements stated in [25]. In particular, for large x , $s(x)$ should monotonically decrease as $O(x^{-1})$ to reduce the impact of those $\tilde{\mathbf{y}}_k$ that do not fit well the average spatial structure of data described by $\tilde{\mathbf{X}}$. In fact, through (25), a large $|\tilde{\mathbf{y}}_k|$ indicates the likely presence of either a strong leakage of the k th focused eigenvector into the error subspace of $\mathbf{Z}$, originated by mismodeling, or an undetected narrowband source. In both cases, the sensitivity of WAVES is minimized by reducing the eigenvector weight [25].

Even if the optimization of $s(x)$ is left as an open topic of research,⁵ a remarkable statistical performance, coupled with a fast and safe numerical convergence, has been obtained by choosing

$$s(x) = \left(x + a \sqrt{M/\eta} \right)^{-1}. \quad (32)$$

For high values of $|\tilde{\mathbf{y}}_k|$, $s(|\tilde{\mathbf{y}}_k|)|\tilde{\mathbf{y}}_k| \cong 1$, limiting the influence of each weighted eigenvector on the WAVES estimate. Formula (32) also limits the weight of those $\tilde{\mathbf{y}}_k$ that are too close to the origin (i.e., $|\tilde{\mathbf{y}}_k| \cong 0$) to furnish a reliable directional information. Such observations usually arise from wrongly selected noise eigenvectors.

The scalar $a > 0$ bounds the maximum weight with respect to the size of $\tilde{\mathbf{Z}}$. In particular, $a = 0.05$ was adopted during simulations, but this choice was not critical for the final performance.

D. Eigenvector Selection

A potentially critical issue in the WAVES approach is the selection of signal eigenvectors to be included in $\tilde{\mathbf{Z}}$. Information theoretic criteria, such as AIC or MDL [23], could be used for this purpose, but they are computationally expensive and nonrobust since they have been derived from asymptotical expansions of the Gaussian likelihood.

A simple and computationally inexpensive scheme was adopted during all experiments, assuming a *contaminated* [25] Gaussian or χ^2 distribution for sample noise eigenvalues. The median md and the robust sample dispersion⁶ $mstd$ of the set of the smallest SCM eigenvalues

$$\left\{ \lambda_k(f_i); k = \frac{M}{2} + 1, \dots, M; i = 1, 2, \dots, J \right\}$$

⁵The “distribution” of x [25], [32] is clearly ill-defined in the context of the WAVES concept.

⁶Based on the median absolute deviation, as suggested in [25].

were computed. If an eigenvalue exceeded the threshold $t = md + 3 \cdot mstd$, the corresponding eigenvector was inserted into $\tilde{\mathbf{Z}}$.

This *ad hoc* scheme, which is currently under development and analysis, performed reliably in simulations. Some noise subspace eigenvectors were erroneously included in $\tilde{\mathbf{Z}}$ at high SNR, apparently because finite bandwidth effects raised the corresponding eigenvalues significantly over the background noise level [13]. At low SNR, instead, some signal eigenvectors might be discarded, thus generating a threshold effect in DOA estimation. However, these mistakes were effectively tackled by the proposed robust WAVES processing so that the global performance was not noticeably impaired.

E. Eigenvector Whitening

Nonunitary focusing may alter the sphericity of the error subspace because (21) shows that sample eigenvector errors are transformed exactly as spatial noise components in (5). In particular, (10) and (17) show that finite sample errors can be treated like i.i.d. noise samples for $N \gg M$.⁷ If these errors dominate, a whitened pseudodata matrix can be redefined from (6) as $\tilde{\mathbf{Z}}_{\mathbf{Q}} \triangleq \mathbf{Q}^{-1}\tilde{\mathbf{Z}}$. Obviously, the same $\mathbf{Q}$ must be used for DOA estimation. However, for a sufficiently large N , (26) is certainly satisfied, and *Theorem 2* states that even the choice $\mathbf{Q} = \mathbf{I}_M$ leads to a consistent WAVES estimate.

VII. ENHANCED SYNTHESIS OF FOCUSING MATRICES

Synthesis of focusing matrices plays a fundamental role in optimizing the statistical performance of CSSM [17], [19] and WAVES. In particular, BI-CSSM fitting (8) is performed for angles $\{\theta_l; l = 1, 2, \dots, L\}$ at each frequency f_i .

Perfect focusing (4) is approximated through a proper design of the reference beamforming matrix $\mathbf{T}_0$, using $K < M$ reference beamformers [18]. In this case, $\tilde{\mathbf{Z}}$ has size $K \times \eta$.

It is common practice to solve (8) in the LS sense. However, computer experiments show that the LS criterion may not be optimal in reducing the frequency-dependent spread among *focused* steering vectors, which increases bias and variance of DOA estimates.

Statistical performance of narrowband subspace algorithms mainly depends on eigenvector perturbations along steering vector derivatives with respect to the P parameters $\{\phi_p; p = 1, 2, \dots, P\}$ collected in vector θ [9], [11], [31].

An enhanced weighted LS criterion is proposed, which explicitly takes into account focusing errors in the derivative subspace. This criterion includes an angle-dependent scalar weighting and minimizes the functional

$$F(f_i) = \sum_{l=1}^L \beta_{il} \mathbf{t}_{il}^H (\mathbf{I}_K + \mathbf{D}_l \mathbf{\Omega}_{il} \mathbf{D}_l^H) \mathbf{t}_{il} \quad (33)$$

⁷DOA estimation with very few wideband observations can be directly performed in the snapshot-frequency domain. Details are omitted here due to space limitations.

where

$$\begin{aligned} \mathbf{D}_l &= \mathbf{\Pi}(f_0, \theta_l) \mathbf{T}_0 \\ &\cdot \left[\frac{\partial \mathbf{b}(f_0, \theta)}{\partial \phi_1} \Big|_{\theta=\theta_l}, \dots, \frac{\partial \mathbf{b}(f_0, \theta)}{\partial \phi_P} \Big|_{\theta=\theta_l} \right] \\ \mathbf{\Pi}(f_0, \theta_l) &= \mathbf{I}_K - \frac{\mathbf{T}_0 \mathbf{b}(f_0, \theta_l) \mathbf{b}^H(f_0, \theta_l) \mathbf{T}_0^H}{|\mathbf{T}_0 \mathbf{b}(f_0, \theta_l)|_2^2} \\ \mathbf{t}_{il} &= \mathbf{T}_0 \mathbf{b}(f_0, \theta_l) - \mathbf{T}(f_i) \mathbf{a}(f_i, \theta_l). \end{aligned} \quad (34)$$

In (33), each β_{il} is a prefixed positive scalar, and $\mathbf{\Omega}_{il}$ is any $P \times P$ positive definite Hermitian matrix. Derivatives in (34) can be analytically evaluated if the reference array geometry is properly chosen.

Fast minimization of (33) can be carried out by the modified Newton algorithm described in [34] using complex arithmetic. Convergence was reached within 20 iterations in most cases, starting from the unweighted LS solution.

VIII. COMPUTER SIMULATIONS

The WAVES approach was tested by Monte Carlo simulations on the same scenario used in [17] and [19]. An eight-sensor uniform linear array received the signals, radiated by four far-field, uncorrelated, and equipowered white Gaussian sources located at 8° , 13° , 33° , and 37° , referred to the array broadside. The sensor passband was $(60 \div 140)$ Hz. Intersensor spacing was half a wavelength at the focusing frequency $f_0 = 100$ Hz. FFT processing was performed in the band $(80 \div 120)$ Hz, using 33 bins out of 64 and a rectangular window. The SNR was referred to a single sensor. Sample bias and variance were computed on the basis of 250 independent simulations for each SNR value, using $N = 100$ snapshots. ROOT-MUSIC [11] was used for the final DOA estimation, setting the exact number of sources.

A. WAVES versus CSSM With a Priori Beamforming

Figs. 1 and 2 show a performance comparison between WAVES and CSSM, using the classical *rotational signal subspace* (RSS) focusing matrices, built from the angles listed in [17], [19], along with the CRB and the asymptotical WAVES-WSF performance. This experiment aimed to investigate the algorithm behavior in a low-error case since RSS matrices satisfy all optimality requirements listed in [35].

Results were indeed very similar. CSSM scored a better performance at 0 dB SNR but a generally worse bias at higher SNR, except for the *single* source at 33° .

At very low SNR, the predicted performance of WAVES-WSF apparently exceeds the element-space CRB. This effect comes from the assumption of *perfect subspace mapping* [19], which in turn requires the *prior knowledge* of true angles.

B. WAVES versus BI-CSSM Over Wide Angular Sectors

The array was then focused over the nonambiguous angular sector $[-44^\circ \div 44^\circ]$, following the BI-CSSM approach [18], with $L = 85$ angles and $K = 6$ reference beamformers. This experimental setup pointed out the resistance of WAVES to larger errors originated by focusing over wide angular sectors.

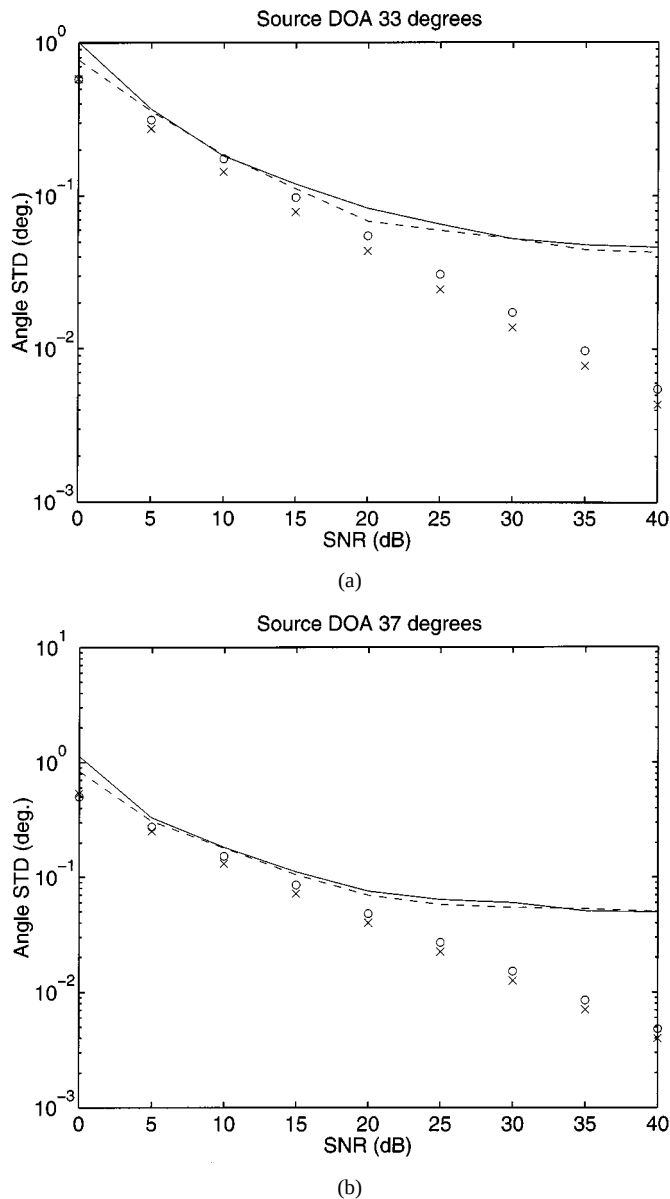


Fig. 1. (a)–(b) Angle STD versus the SNR for sources at 33° and 37°. Solid line: WAVES; dashed line: CSSM; x-marks: CRB; circles: asymptotical WAVES-WSF.

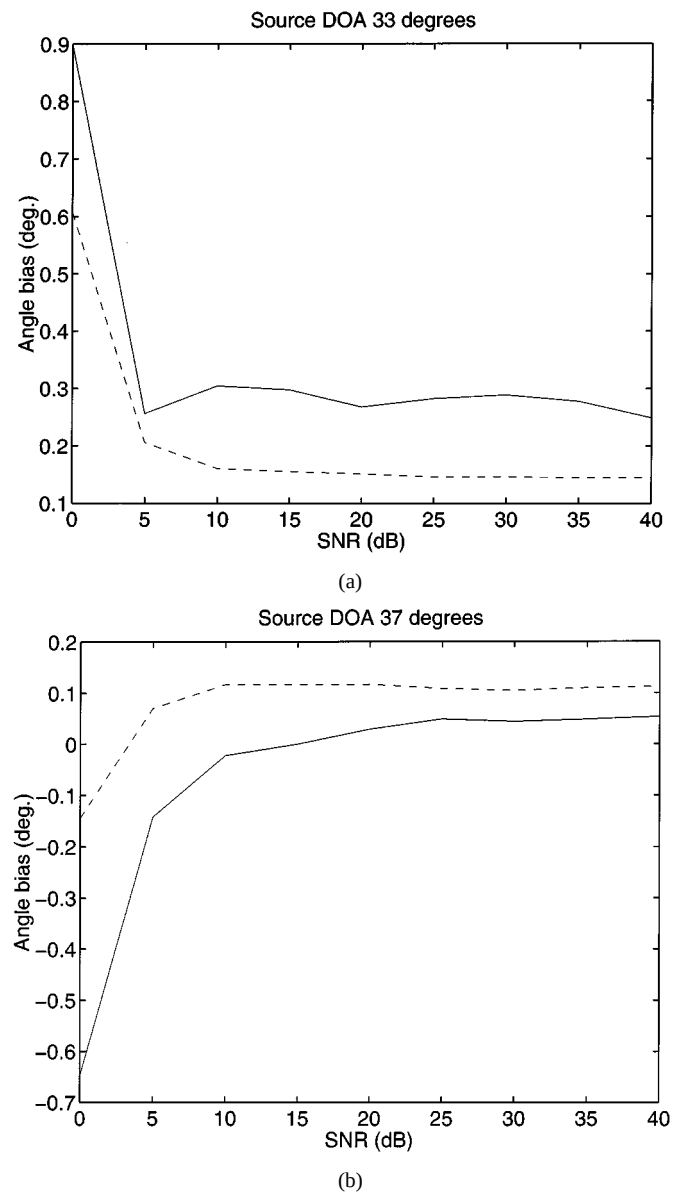


Fig. 2. (a)–(b) Angle bias versus the SNR for sources at 33° and 37°. Solid line: WAVES; dashed line: CSSM.

Results are displayed in Figs. 3 and 4 for the standard BI-CSSM with LS focusing, the BI-CSSM modified with the proposed error functional (33), and two versions of the WAVES approach. The same reference beamforming matrix $\mathbf{T}_0$ and beam-space ROOT-MUSIC [18] were used in all cases.

The two WAVES-based algorithms adopted the new focusing (33), but only the first one exploited eigenvector whitening by the matrix $\mathbf{Q}$. Since focusing matrices were numerically well-conditioned [26], performance was nearly the same in both cases.

Information loss due to beam-space transformations raised the SNR resolution threshold [35] up to about 10 dB for all algorithms.

Adaptive optimization of WAVES led to an appreciable reduction of the estimation variance over the standard BI-CSSM. It is important to remark that in most cases, dominant eigenvectors were underweighted by the robust WAVES estimator, as predicted by theory.

Results are shown for the most difficult angles (33° and 37°). Similar behavior was observed for the other two sources. The tighter bias control obtained by (33) is clearly evident.

Below the resolution threshold, the centroids of the two source clusters were correctly localized. Therefore, another local WAVES or WSF algorithm might be started to refine DOA estimates, exploiting partial results from the first stage, such as the noise power estimate, signal subspaces, and eigenvector weights.

C. Robustness to Narrowband Interference

The third experiment tested the robustness of WAVES to unwanted narrowband interference on wideband DOA estimation.

The two sources at 13° and 33° were considered as interference and generated by AR processes, whose transfer functions were, respectively, $H_{13}(z) = (1 - 0.9e^{j\pi/3}z^{-1})^{-1}$ and

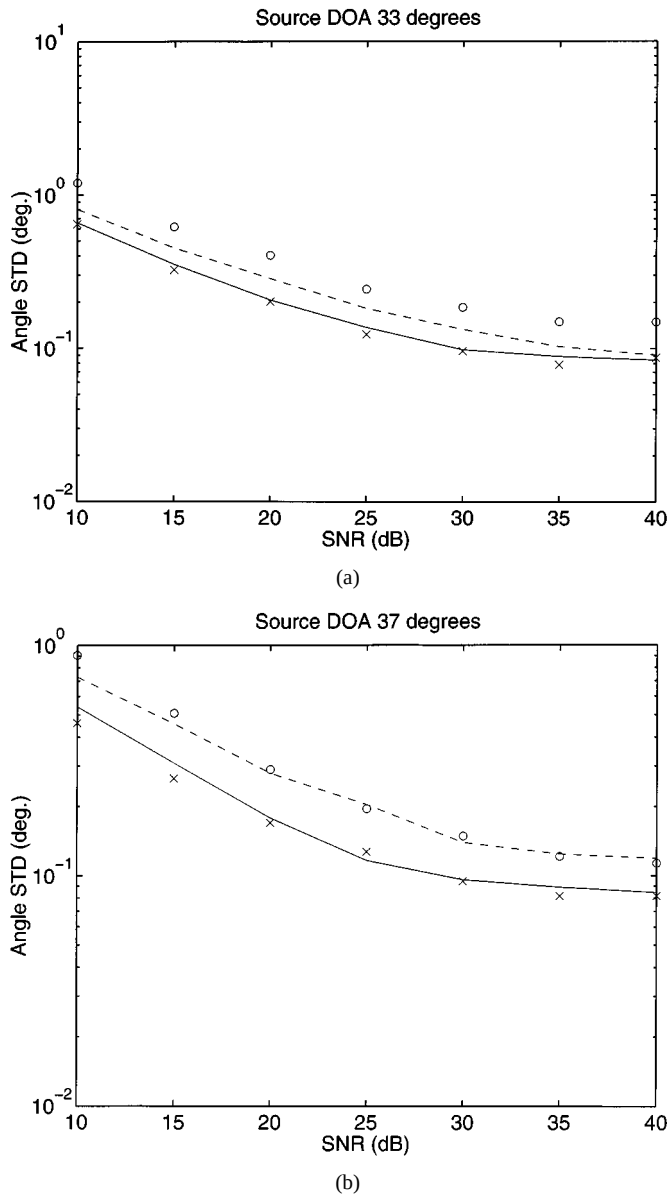


Fig. 3. (a)–(b) Angle standard deviation in degrees versus the SNR for sources at 33° and 37° . Dashed line: BI-CSSM with LS focusing; circles: BI-CSSM using the proposed focusing; solid line: WAVES with the proposed focusing; cross-marks: WAVES with the proposed focusing and $\mathbf{Q} = \mathbf{I}$.

$H_{33}(z) = (1 - 0.9e^{-j\pi/4}z^{-1})^{-1}$. The SNR was redefined with respect to the variance of the white driving noise.

The robust WAVES was compared with the standard BI-CSSM [18]. The same focusing matrices optimized through (33) were employed.

In this case, AR sources were *attenuated* by the robust WAVES algorithm by about 10 dB, as evidenced by both singular value analysis of $\tilde{\mathbf{Z}}$ and DOA standard deviation. In contrast, WAVES markedly improved the BI-CSSM performance for the white sources of interest, placed at 8° and 37° . In particular, Figs. 5 and 6 show the results for the source at 37° .

This time, the preliminary beamforming proposed in [17] and [19] produced unacceptable estimates of cluster centroids that were strongly biased toward interfering sources.

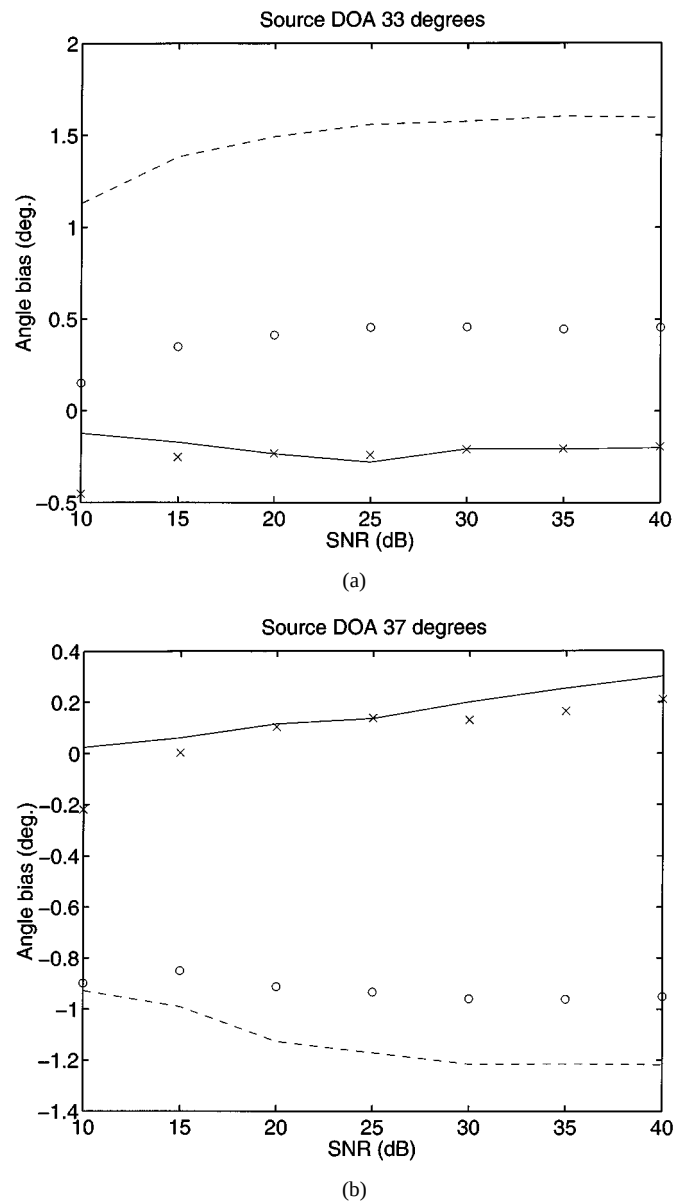


Fig. 4. (a)–(b) Angle bias in degrees versus the SNR for sources at 33° and 37° . Dashed line: BI-CSSM with LS focusing; circles: BI-CSSM with the proposed focusing; solid line: WAVES with the proposed focusing; cross-marks: WAVES with the proposed focusing and $\mathbf{Q} = \mathbf{I}$.

IX. DISCUSSION

The novel WAVES statistic gets inspiration from the asymptotically efficient WSF and optimizes the weighting of focused narrowband signal subspaces. Global results of WAVES-based algorithms are typical of *robust* estimators [25]. A very small performance loss in *ideal* cases is traded off with a strong reduction of the *actual risk* of getting unacceptable DOA estimates in the presence of unbalanced data sets originated by highly varying source spectra, narrowband interference and, unmodeled focusing errors.

As a matter of fact, when coupled with a modified LS criterion for focusing, WAVES significantly *improves* the performance of existing coherent wideband direction-finding algorithms with respect to typical mismodeling, even in the finite-sample case. In particular, experiments confirm the importance

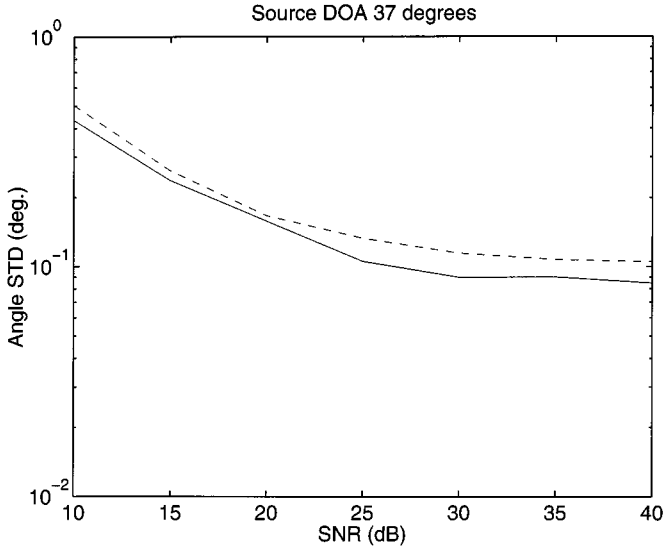


Fig. 5. Angle STD versus the SNR for the white source at 37°. Solid line: WAVES; dashed line: BI-CSSM.

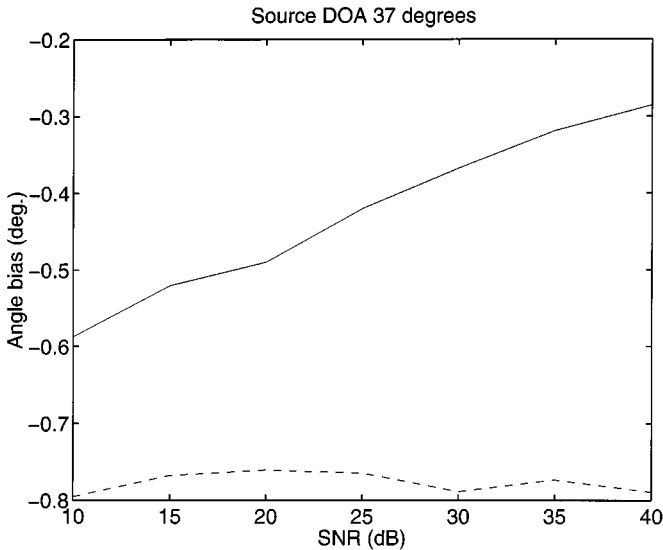


Fig. 6. Angle bias versus the SNR for the white source at 37°. Solid line: WAVES; dashed line: BI-CSSM.

of a sound design of focusing matrices, based on performance considerations to reduce the bias and the angular spread of focused wideband sources.

The performance of WAVES is in close agreement with the theoretical analysis at low SNR. However, the presence of focusing errors raises the DOA variance at high SNR, thus impairing the consistency of *all* CSSM-based estimators.

The computational cost of WAVES is indeed comparable with that of the standard CSSM and strongly depends on the particular implementation. In fact, the WAVES approach allows several design tradeoffs with performance (input filtering, beam-space transformations, eigenvector whitening). The overhead of WAVES algorithms presented here is comparable with the cost of processing about 50 extra wideband snapshots. It is worth pointing out that this cost *does not increase* when acquiring more data in a *stationary* environment. In fact, since $\tilde{\mathbf{R}}_{xx}(f_i)$ is statistically consistent, the number of eigenvectors to be processed in $\tilde{\mathbf{Z}}$ is almost independent of N . The main effort is the

computation of many small-sized QRD and SVD that can be effectively parallelized on modern DSP multiprocessor machines [26], [29], [30].

APPENDIX A

ASYMPTOTICAL CONVERGENCE OF THE MATRIX $\tilde{\mathbf{Z}}$

As shown in Section III, each $M \times \eta_i$ block $\tilde{\mathbf{Z}}_i$ of the matrix $\tilde{\mathbf{Z}}$ verifies

$$\|\tilde{\mathbf{Z}}_i - \mathbf{Z}_i\|_2 \rightarrow 0 \quad \text{w.p. 1.}$$

It is also shown here that $\|\mathbf{E}\|_2 = \|\sqrt{\eta}(\tilde{\mathbf{Z}} - \mathbf{Z})\|_2 \rightarrow 0$ w.p. 1. In fact, if $\mathbf{u}$ and $\mathbf{v}$ are, respectively, the principal left and right singular vectors of $\mathbf{E}$, then $\|\mathbf{E}\|_2 = \sqrt{\eta}|\mathbf{u}^H(\tilde{\mathbf{Z}} - \mathbf{Z})\mathbf{v}|_2$ [29]. If $\mathbf{v}$ is partitioned as

$$\mathbf{v}^H = [\mathbf{v}_1^H, \dots, \mathbf{v}_J^H]$$

it follows that

$$\begin{aligned} \|\mathbf{E}\|_2 &\leq \sum_{i=1}^J |\mathbf{u}^H(\tilde{\mathbf{Z}}_i - \mathbf{Z}_i)\mathbf{v}_i|_2 \leq \sum_{i=1}^J \|\tilde{\mathbf{Z}}_i - \mathbf{Z}_i\|_2 \|\mathbf{v}_i\|_2 \\ &\leq \max_{1 \leq i \leq J} \|\tilde{\mathbf{Z}}_i - \mathbf{Z}_i\|_2 \cdot \left(\sum_{i=1}^J \|\mathbf{v}_i\|_2 \right) \rightarrow 0 \quad \text{w.p. 1.} \end{aligned}$$

APPENDIX B

SVD-BASED ALGORITHM FOR SOLVING (31)

Solution of (31) has been approached in [25] by iterative Cholesky factorization, which may be numerically unstable, and other iterative approaches, similar to those described in [29] and [30]. However, no algorithm was recognized at the time to be sufficiently fast and reliable for real-time applications. Therefore, a novel iterative technique has been devised, based on the SVD [26].

Without loss of generality, it is supposed that the starting $m \times n$ complex data matrix $\mathbf{Y}$ has more rows (containing system observations, [6]) than columns. The algorithm can be summarized as follows.

Initialization: Set the iteration counter $q = 0$ and compute the R-SVD $\mathbf{Y} = \mathbf{U}_{(0)}\Sigma_{(0)}\mathbf{W}_0^H$. Save the “tiny” $m \times n$ orthogonal matrix $\mathbf{U}_{(q)} \equiv \mathbf{U}_{(0)}$, and set $\mathbf{X}_{(q)} = \mathbf{W}_{(0)}\Sigma_{(0)}$.⁸

Step 1: For $k = 1, 2, \dots, m$, compute row (observation) weights $\mathbf{S}_{(q)[k,k]}$ through (32), as⁹

$$r_{(q)k} = \|\mathbf{U}_{(q)}(k, 1:n)\|_2$$

$$\mathbf{S}_{(q)[k,k]} = s(r_{(q)k}) \left[\frac{1}{m} \sum_{j=1}^m s(r_{(q)j})^2 \right]^{-1/2}. \quad (35)$$

These weights are formally stored in the diagonal $m \times m$ matrix $\mathbf{S}_{(q)}$.

⁸In order to avoid possible numerical artifacts, subsequent computations should be carried only on non-negligible components of $\mathbf{Y}$ revealed from its SVD [26]. Details of this rank-reduced algorithm are omitted here.

⁹This formula scales the matrix $\mathbf{X}$ as the standard SVD does. The proper covariance scaling is obtained by removing the dependence on m .

Step 2: Compute factors $\Sigma_{(q+1)}$ and $\mathbf{W}_{(q+1)}$ of the R-SVD $\mathbf{S}_{(q)}\mathbf{U}_{(q)} = \mathbf{V}_{(q+1)}\Sigma_{(q+1)}\mathbf{W}_{(q+1)}^H$.

Step 3: Update $\mathbf{X}_{(q+1)} = \mathbf{X}_{(q)}\mathbf{W}_{(q+1)}\Sigma_{(q+1)}$ and $\mathbf{U}_{(q+1)} = \mathbf{U}_{(q)}\mathbf{W}_{(q+1)}\Sigma_{(q+1)}^{-1}$. The last operation is usually well conditioned since the matrix $\mathbf{S}_{(q)}\mathbf{U}_{(q)}$ should be nearly orthogonal after the preprocessing.

Step 4: If

$$\sum_{j=1}^n |\Sigma_{(q+1)}[j, j] - 1| < tol \cdot n$$

where tol is a small constant (e.g., 10^{-4}), set $\mathbf{X} = \mathbf{X}_{(q+1)}$ and exit. Otherwise, set $q = q + 1$, and return to Step 1.

The resulting $n \times n$ matrix $\mathbf{X}$ is a (scaled) square-root of the sample pseudocovariance. Convergence was usually achieved within about five iterations if $\mathbf{Y}$ contained identically distributed Gaussian data. More iterations ($10 \div 20$) were required to compute the WAVES during computer simulations described above. Interestingly, the slower convergence at high SNR confirms the influence of focusing errors. The rate was further reduced when dealing with data contaminated by outliers [25], but the algorithm always converged.

REFERENCES

- [1] H. Krim and M. Viberg, "Two decades of array signal processing research," *IEEE Signal Processing Mag.*, pp. 67–94, July 1996.
- [2] S. Haykin, Ed., *Array Signal Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1984.
- [3] G. Xu, H. P. Lin, S. S. Jeng, and W. J. Vogel, "Experimental studies of spatial signature variation at 900 mhz for smart antenna systems," *IEEE Trans. Antennas Propagat.*, vol. 46, pp. 953–962, July 1998.
- [4] R. O. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Trans. Antennas Propagat.*, vol. AP-34, pp. 276–280, Mar. 1986.
- [5] G. Bienvenu and L. Kopp, "Optimality of high-resolution array processing using the eigensystem approach," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. ASSP-31, pp. 1235–1248, Oct. 1983.
- [6] B. De Moor, "The singular value decomposition and long and short spaces of noisy matrices," *IEEE Trans. Signal Processing*, vol. 41, pp. 2826–2838, Sept. 1993.
- [7] R. Roy and K. Kailath, "Esprit—Estimation of signal parameter via rotational invariance techniques," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. 37, pp. 984–995, July 1989.
- [8] R. Kumaresan and D. W. Tufts, "Estimating the angles of arrival of multiple plane waves," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-19, pp. 134–139, Jan. 1983.
- [9] M. Viberg, B. Ottersten, and T. Kailath, "Detection and estimation in sensor arrays using weighted subspace fitting," *IEEE Trans. Signal Processing*, vol. 39, pp. 2436–2449, Nov. 1991.
- [10] P. Stoica and A. Nehorai, "Music, maximum likelihood, and Cramer–Rao bound," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. 37, pp. 720–741, May 1989.
- [11] B. D. Rao and K. V. S. Hari, "Performance analysis of root-music," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. 37, pp. 1939–1949, Dec 1989.
- [12] J. Krolik and D. N. Swingler, "Multiple wide-band source location using steered covariance matrices," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. 37, pp. 1481–1494, Oct. 1989.
- [13] M. Agrawal and S. Prasad, "Broadband doa estimation using spatial-only modeling of array data," *IEEE Trans. Signal Processing*, vol. 48, pp. 663–670, Mar. 2000.
- [14] H. Wang and M. Kaveh, "Coherent signal-subspace processing for the detection and estimation of angles of arrival of multiple wide-band sources," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. ASSP-33, pp. 823–831, Aug. 1985.
- [15] M. A. Doron, A. J. Weiss, and H. Messer, "Maximum-likelihood direction finding of wide-band sources," *IEEE Trans. Signal Processing*, vol. 41, pp. 411–414, Jan. 1993.
- [16] J. A. Cadzow, "Multiple source location—The signal subspace approach," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. 38, pp. 1110–1125, July 1990.
- [17] H. Hung and M. Kaveh, "Focusing matrices for coherent signal-subspace processing," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. 36, pp. 1272–1281, Aug. 1988.
- [18] T.-S. Lee, "Efficient wide-band source localization using beamforming invariance technique," *IEEE Trans. Signal Processing*, vol. 42, pp. 1376–1387, June 1994.
- [19] M. A. Doron and A. J. Weiss, "On focusing matrices for wide-band array processing," *IEEE Trans. Signal Processing*, vol. 40, pp. 1295–1302, June 1992.
- [20] B. Friedlander and A. J. Weiss, "Direction finding for wide-band signals using an interpolated array," *IEEE Trans. Signal Processing*, vol. 41, pp. 1618–1635, Apr. 1993.
- [21] A. L. Swindlehurst and T. Kailath, "A performance analysis of subspace-based methods in the presence of model errors—Part I: The music algorithm," *IEEE Trans. Signal Processing*, vol. 40, pp. 1758–1774, July 1992.
- [22] —, "A performance analysis of subspace-based methods in the presence of model errors—Part II: Multidimensional algorithms," *IEEE Trans. Signal Processing*, vol. 41, pp. 2882–2890, Sept. 1993.
- [23] M. Wax and T. Kailath, "Detection of signals by information theoretic criteria," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. ASSP-33, pp. 387–392, Apr. 1985.
- [24] S. Valaee and P. Kabal, "Wide-band array processing using a two-sided correlation transformation," *IEEE Trans. Signal Processing*, vol. 43, pp. 160–172, Jan. 1995.
- [25] P. J. Huber, *Robust Statistics*. New York: Wiley, 1981.
- [26] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 2nd ed. Baltimore, MD: John Hopkins Univ. Press, 1989.
- [27] M. A. Doron and E. Doron, "Wavefield modeling and array processing—Part II: Algorithms," *IEEE Trans. Signal Processing*, vol. 42, pp. 2560–2570, Oct. 1994.
- [28] S. U. Pillai and C. S. Burrus, *Array Signal Processing*. New York: Springer-Verlag, 1989.
- [29] P. Strobach, "Bi-iteration svd subspace tracking algorithms," *IEEE Trans. Signal Processing*, vol. 45, pp. 1222–1240, May 1997.
- [30] —, "Fast orthogonal iteration adaptive algorithms for the generalized symmetric eigenproblem," *IEEE Trans. Signal Processing*, vol. 46, pp. 3345–3359, Dec. 1998.
- [31] A. J. Weiss and B. Friedlander, "Effects of modeling errors on the resolution threshold of the music algorithm," *IEEE Trans. Signal Processing*, vol. 42, pp. 1519–1526, June 1994.
- [32] D. B. Williams and D. H. Johnson, "Robust estimation of structured covariance matrices," *IEEE Trans. Signal Processing*, vol. 41, pp. 2891–2906, Sept. 1993.
- [33] D. G. Luenberger, *Linear and Nonlinear Programming*, 2nd ed. Reading, MA: Addison-Wesley, 1989.
- [34] R. Parisi, E. D. Di Claudio, G. Orlandi, and B. D. Rao, "A generalized learning paradigm exploiting the structure of feedforward neural networks," *IEEE Trans. Neural Networks*, vol. 7, pp. 1450–1460, Nov. 1996.
- [35] A. J. Weiss and B. Friedlander, "Preprocessing for direction finding with minimal variance degradation," *IEEE Trans. Signal Processing*, vol. 42, pp. 1478–1485, June 1994.



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